University of Prishtina "Hasan PRISHTINA"

Faculty of Electrical and Computer Engineering
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Course: Data Preparation and Visualization

Week IV – Data Preprocessing II

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What is today's agenda?

➢ **Today we are going to discuss about:**

- Data Preprocessing;
- Data Cleaning;
- Data Integration.

Recap: Data Preprocessing

- Data Preprocessing: An Overview
  - Data Quality
  - Major Tasks in Data Preprocessing
- Data Cleaning
- Data Integration

Data Quality: Why Preprocess the Data?

- Measures for data quality, a multidimensional view:
  - **Accuracy**: correct or wrong, accurate or not;
  - **Completeness**: not recorded, unavailable, …
  - **Consistency**: some modified but some not, dangling, …
  - **Timeliness**: timely update?
  - **Believability**: how trustable the data are correct?
  - **Interpretability**: how easily the data can be understood?

Major Tasks in Data Preprocessing

- **Data cleaning:**
  - Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies.
- **Data integration:**
  - Integration of multiple databases, data cubes, or files.
- **Data reduction:**
  - Dimensionality reduction;
  - Numerosity reduction;
  - Data compression.
- **Data transformation and data discretization:**
  - Normalization;
  - Concept hierarchy generation.

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Data Cleaning

- Data in the Real World Is Dirty: Lots of potentially incorrect data, e.g., instrument faulty, human or computer error, transmission error.
  - incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data:
    - e.g., *Occupation*="" (missing data)
  - noisy: containing noise, errors, or outliers:
    - e.g., *Salary*="-10" (an error)
  - inconsistent: containing discrepancies in codes or names, e.g.:
    - *Age*="42", *Birthday*="03/07/2010"
    - Was rating "1, 2, 3", now rating "A, B, C"
    - discrepancy between duplicate records
  - Intentional (e.g., *disguised missing* data):
    - Jan. 1 as everyone's birthday?

Incomplete (Missing) Data

- Data is not always available:
  - E.g., many tuples have no recorded value for several attributes, such as customer income in sales data.
- Missing data may be due to:
  - equipment malfunction;
  - inconsistent with other recorded data and thus deleted;
  - data not entered due to misunderstanding;
  - certain data may not be considered important at the time of entry;
  - not register history or changes of the data.
- Missing data may need to be identified.

How to Handle Missing Data?

- **Ignore the tuple**:

  usually done when class label is missing (when doing classification) - not effective when the % of missing values per attribute varies considerably;

- Fill in the missing value **manually**: tedious + infeasible?
- Fill in it **automatically** with:
  - a global constant: e.g., “unknown”, a new class?!
  - the attribute mean;
  - the attribute mean for all samples belonging to the same class: **smarter**;
  - the most probable value: inference-based such as Bayesian formula or decision tree.

Noisy Data

- **Noise**: random error or variance in a measured variable.
- **Incorrect attribute** values may be due to:
  - faulty data collection instruments;
  - data entry problems;
  - data transmission problems;
  - technology limitation;
  - inconsistency in naming convention.
- Other **data problems** which require data cleaning:
  - duplicate records;
  - incomplete data;
  - inconsistent data.

How to Handle Noisy Data?

- **Binning:**
  - first sort data and partition into (equal-frequency) bins;
  - then one can smooth by bin means, smooth by bin median, smooth by bin boundaries, etc.
- **Regression:**
  - smooth by fitting the data into regression functions;
- **Clustering:**
  - detect and remove outliers;
- **Combined computer and human inspection:**
  - detect suspicious values and check by human (e.g., deal with possible outliers).

Data Cleaning as a Process

➢ Data discrepancy detection:

- Use metadata (e.g., domain, range, dependency, distribution);
- Use commercial tools:
  - **Data scrubbing**: use simple domain knowledge (e.g., postal code, spell-check) to detect errors and make corrections;
  - **Data auditing**: by analysing data to discover rules and relationship to detect violators (e.g., correlation and clustering to find outliers).

Data Cleaning as a Process

- Data **migration** and **integration**:
  - Data **migration** tools: allow **transformations** to be specified;
  - **ETL** (Extraction/Transformation/Loading) tools: allow users to specify transformations through a graphical user interface.
- Integration of the two processes:
  - Iterative and interactive (e.g., Potter's Wheels)

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Data Integration

- **Data integration**:
  - Combines data from multiple sources into a coherent store;
- Schema **integration**: e.g., A.cust-id ≡ B.cust-#:
  - Integrate metadata from different sources.
- Entity **identification** problem:
  - Identify real world entities from multiple data sources, e.g., Bill Clinton = William Clinton.
- **Detecting** and **resolving** data value conflicts:
  - For the same real-world entity, attribute values from different sources are different;
  - **Possible reasons**: different representations, different scales, e.g., metric vs. British units.

Handling Redundancy in Data Integration

- **Redundant data** occur often when integration of multiple databases:
  - **Object identification**: The same attribute or object may have different names in different databases;
  - **Derivable data**: One attribute may be a "derived" attribute in another table, e.g., annual revenue.
- **Redundant attributes** may be able to be detected by correlation analysis and covariance analysis;
- Careful integration of the data from multiple sources may help reduce/avoid redundancies and inconsistencies and improve mining speed and quality.

Correlation Analysis (Nominal Data)

- **X^2 (chi-square) test**

$$\chi^2 = \sum \frac{(Observed - Expected)^2}{Expected}$$

- The larger the X^2 value, the more likely the variables are related;
- The cells that contribute the most to the X^2 value are those whose actual count is very different from the expected count;
- Correlation does not imply causality:
  - # of hospitals and # of car-theft in a city are correlated;
  - Both are causally linked to the third variable: population.

Chi-Square Calculation: An Example

	Play Chess	Not play chess	Sum (row)
Like science fiction	250 (90)	200 (360)	450
Not like science fiction	50 (210)	1000 (840)	1050
Sum (col.)	300	1200	1500

$$\chi^2 = \sum \frac{(Observed - Expected)^2}{Expected}$$

- X^2 (chi-square) calculation (numbers in parenthesis are expected counts calculated based on the data distribution in the two categories):

$$X^2 = \frac{(250 - 90)}{90} + \frac{(50 - 210)}{210} + \frac{(200 - 360)}{360} + \frac{(1000 - 840)}{840} = 507.93$$

- It shows that like_science_fiction and play_chess are correlated in the group.

Exercises

	Attend Tutoring	Do not attend tutoring	Sum (row)
Passed Exam	95 (55)	10 (5)	105
Failed Exam	49 (10)	10 (10)	59
Sum (col.)	144	20	164

$$\chi^2 = \sum \frac{(Observed - Expected)^2}{Expected}$$

Correlation Analysis (Numeric Data)

- Correlation coefficient (also called Pearson's product moment coefficient):

$$r_{A,B} = \frac{\sum_{i=1}^{n}(a_i - \overline{A})(b_i - \overline{B})}{(n-1)\sigma_A \sigma_B} = \frac{\sum_{i=1}^{n}(a_i b_i) - n\overline{A}\overline{B}}{(n-1)\sigma_A \sigma_B}$$

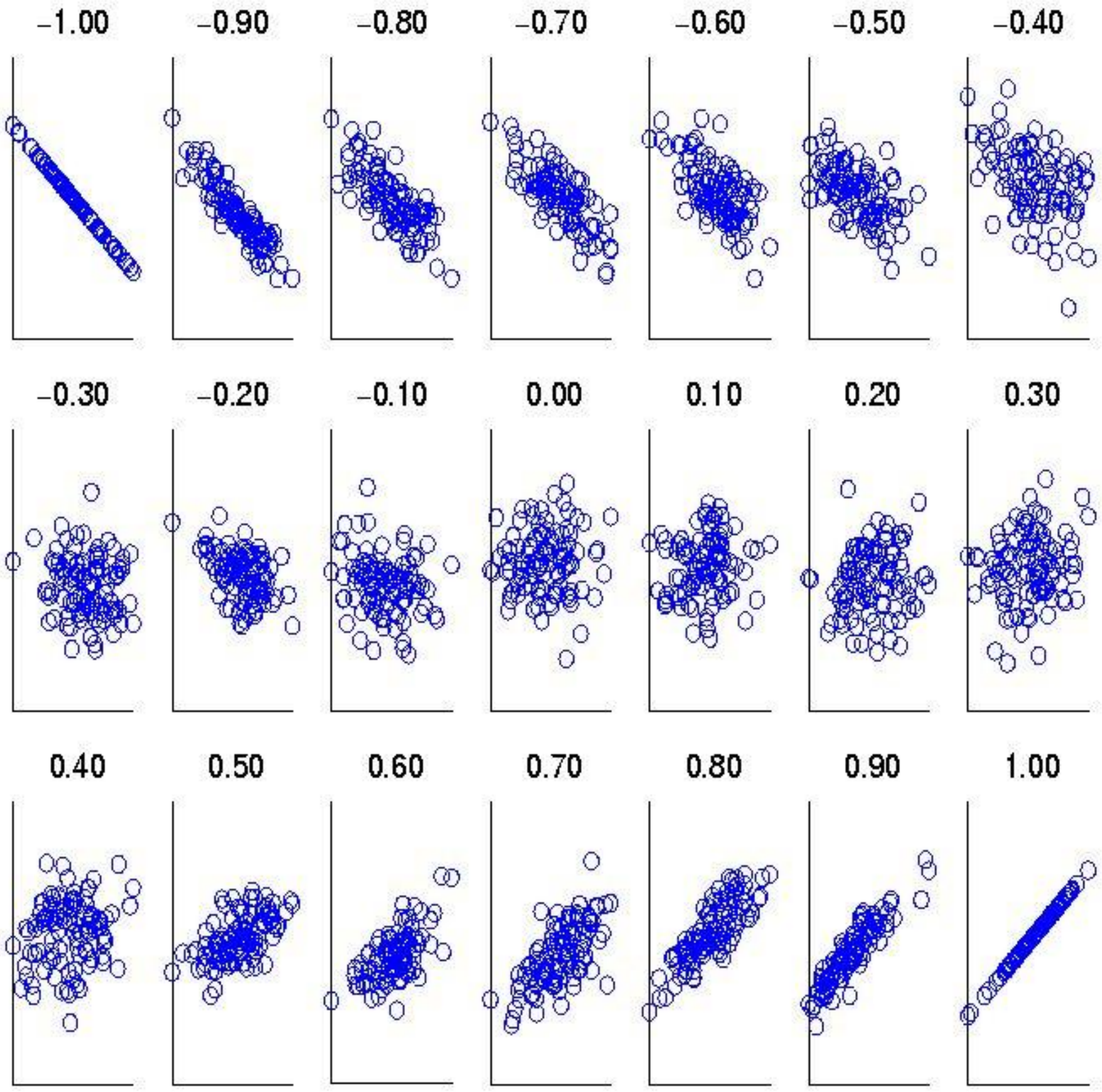
where n is the number of tuples, A and B are the respective means of A and B, σ_A σ_B are the respective standard deviation of A and B, and $\Sigma(a_i b_i)$ is the sum of the AB cross-product.

- If $r_{A,B} > 0$, A and B are positively correlated (A's values increase as B's). The higher, the stronger correlation.

  $r_{A,B} = 0$: independent

  $r_{AB} < 0$: negatively correlated.

Visually Evaluating Correlation



Scatter plots showing the similarity from –1 to 1

Figure 5.11. Scatter plots illustrating correlations from -1 to 1.

Correlation (viewed as linear relationship)

- Correlation measures the linear relationship between objects;
- To compute correlation, we standardize data objects, A and B, and then take their dot product.

$$a'_k = (a_k - mean(A)) / std(A)$$

$$b'_k = (b_k - mean(B)) / std(B)$$

Covariance (Numeric Data)

- Covariance is similar to correlation:

$$Cov(A,B) = E((A-\bar{A})(B-\bar{B})) = \frac{\sum_{i=1}^{n}(a_i-\bar{A})(b_i-\bar{B})}{n}$$

- Correlation coefficient: $r_{A,B} = \dfrac{Cov(A,B)}{\sigma_A \sigma_B}$

A and B are the respective mean or **expected values** of A and B, σ_A and σ_B are the respective standard deviation of A and B.

Covariance (Numeric Data) - I

- **Positive covariance**: If Cov $_{A,B} > 0$, then A and B both tend to be larger than their expected values;

- **Negative covariance**: If Cov $_{A,B} < 0$, then if A is larger than its expected value, B is likely to be smaller than its expected value;

- **Independence**: Cov A,B = 0 but the converse is not true:
  - Some pairs of random variables may have a covariance of 0 but are not independent;
  - Only under some additional assumptions (e.g., the data follow multivariate normal distributions) does a covariance of 0 imply independence.

Covariance: An Example

$$Cov(A,B) = E((A-\bar{A})(B-\bar{B})) = \frac{\sum_{i=1}^{n}(a_i-\bar{A})(b_i-\bar{B})}{n}$$

- It can be simplified in computation as:

$$Cov(A,B) = E(A \cdot B) - \bar{A}\bar{B}$$

- Suppose two stocks A and B have the following values in one week:

(2, 5), (3, 8), (5, 10), (4, 11), (6, 14);

Covariance: An Example - I

➢ Question: If the stocks are affected by the same industry trends, will their prices rise or fall together?

(2, 5), (3, 8), (5, 10), (4, 11), (6, 14)

- $E(A) = (2 + 3 + 5 + 4 + 6)/ 5 = 20/5 = 4$
- $E(B) = (5 + 8 + 10 + 11 + 14) /5 = 48/5 = 9.6$
- $Cov (A,B) = (2\times5+3\times8+5\times10+4\times11+6\times14)/5 - 4 \times 9.6 = 4$

➢ Thus, A and B rise together since $Cov(A, B) > 0$

Exercise

➢ Question: If the stocks are affected by the same industry trends, will their prices rise or fall together?

(4,7), (5,9), (6,12), (7,13), (8,15)

- E(A) = (4 + 5 + 6 + 7 + 13)/ 5 = 30/5 = 6
- E(B) = (7 + 9 + 12 + 13 + 15) /5 = 56/5 = 11.2
- Cov (A,B) = (4×7+5×9+6×12+7×13+8×15)/5 - 6 × 11.2 = 4

➢ Since Cov (A,B) = 4 > 0, t indicates that sales for Product A and Product B rise together as they are positively correlated.

Questions?

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